

Systems of Linear Equations

ANSWER KEY



Setting up the augmented matrix

- 1 Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$ (give it as its three rows).
Row 1: $\langle 2, 1, -1, 3 \rangle$. Row 2: $\langle 1, -3, 2, -1 \rangle$. Row 3: $\langle 4, 1, 1, 5 \rangle$ (each row lists the x, y, z coefficients, then the right-hand side).
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Elementary row operations

- 2 List the three types of elementary row operations.
1. Swap two rows. 2. Multiply a row by a nonzero scalar. 3. Add a multiple of one row to another row.
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Solving 2x2 systems by elimination

- 3 Solve $2x + 3y = 7$, $x - y = 1$.
From the second equation, $x = 1 + y$. Substituting: $2(1 + y) + 3y = 7 \Rightarrow 5y = 5 \Rightarrow y = 1$, so $x = 2$.
Solution: $(2, 1)$.
- 4 Solve $3x - 2y = 4$, $5x + y = 11$.
From the second equation, $y = 11 - 5x$. Substituting: $3x - 2(11 - 5x) = 4 \Rightarrow 13x = 26 \Rightarrow x = 2$, so $y = 1$. Solution: $(2, 1)$.
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Solving 3x3 systems by Gaussian elimination

- 5 Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.
From the first equation, $x = 6 - y - z$. Substituting into the third:
 $2(6 - y - z) + 5y - z = 27 \Rightarrow 3y - 3z = 15 \Rightarrow y = 5 + z$. Substituting into the second:
 $2(5 + z) + 5z = -4 \Rightarrow 7z = -14 \Rightarrow z = -2$, so $y = 3$ and $x = 6 - 3 - (-2) = 5$. Solution: $(5, 3, -2)$.
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Why row operations preserve the solution set

- 6 Explain briefly why multiplying a row by a nonzero constant C doesn't change a system's solution set.
Multiplying both sides of an equation by a nonzero C produces an equivalent equation, since the operation is invertible — dividing by C recovers the original equation exactly.

7 Why is the restriction $c \neq 0$ needed for this row operation?

If $c = 0$, the row becomes the zero row, which is not invertible — the original row's information is destroyed, so the resulting system need not have the same solution set as the original.